

VLSI Design and advanced semiconductor technology

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- Year/Semester: 2nd year /4th semester

Very Large Scale Integration (VLSI) design plays a crucial role in the development of modern electronic systems by enabling the integration of millions to billions of transistors on a single semiconductor chip. With continuous scaling of device dimensions, advanced semiconductor technologies have emerged to improve performance, power efficiency, and reliability of integrated circuits. This abstract presents an overview of VLSI design principles, including design hierarchy, CMOS technology, and key challenges such as power dissipation, interconnect delay, and process variability. It also highlights recent advancements in semiconductor technology, including FinFETs, Gate-All-Around (GAA) transistors, extreme ultraviolet (EUV) lithography, and three-dimensional (3D) integration. These innovations support high-speed, low-power, and high-density circuit design required for applications in processors, memory devices, artificial intelligence, and system-on-chip (SoC) architectures. The combination of VLSI design methodologies and advanced semiconductor technologies continues to drive progress in the electronics industry.